(https://www.dfrobot.com/product-1601.html)

Introduction

This is an industrial incremental photoelectric rotary encoder with aluminum material, metal shell and stainless steel shaft. It generates AB two-phase orthogonal pulse signal though the rotation of the grating disk and optocoupler. 400 pulses/round for each phase, and 1600 pulses/round for dual-phase 4 times output. This rotary encoder supports max 5000 r/min speed. And it can be used for speed, angle, angular velocity and other data measurement. The photoelectric rotary encoder has a NPN open collector output. It could work with Microcontroller with internal pull-up resistors directly. And it is using 750L05 voltage regulator chip, which has a DC4.8V-24V wide range power input, compatible with Arduino, STM32, PLC and other types of microcontrollers.

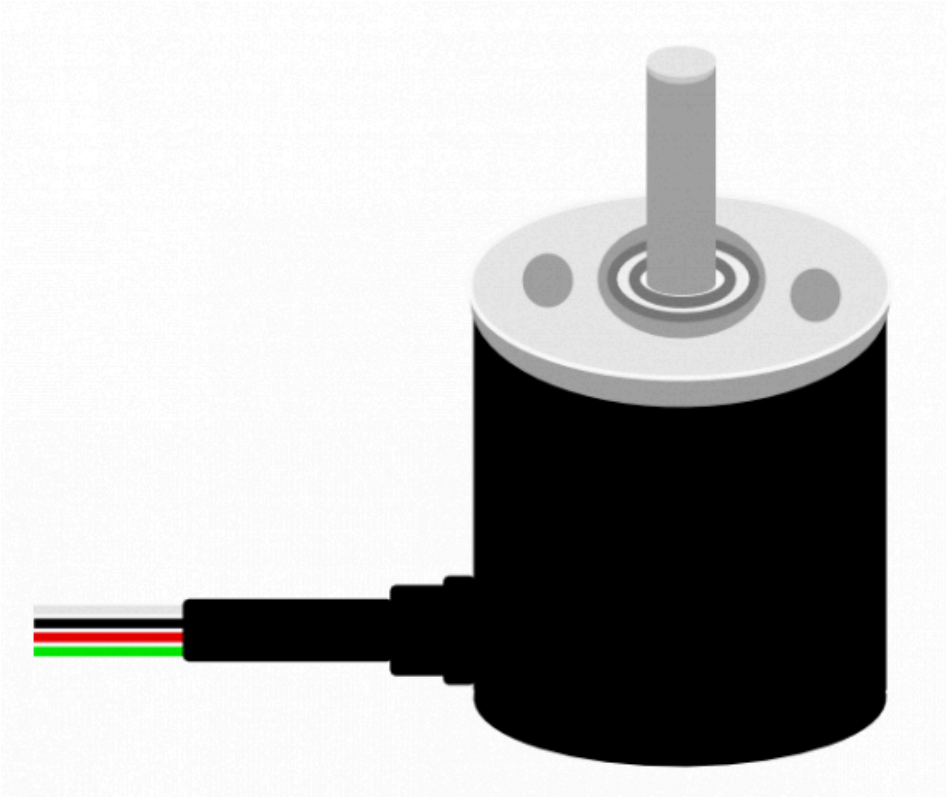
Note: NPN open collector output needs pull-up resistors for the oscilloscope display.

Specification

- Supply Voltage: 4.8V ~ 24v
- Encoder Body Size: Φ39× 36.5mm
- Output Shaft Diameter: Φ6 × 13mm

- Outside Shaft Platform: Φ20 × 4.85 mm
- Fixing Holes Screws: M3

Board Overview



Num	Label	Description
Red	VCC	Power +
Black	GND	Power -
White	A	Pulse A (Need pull-up Resistor)
Green	B	Pulse B (Need pull-up Resistor)

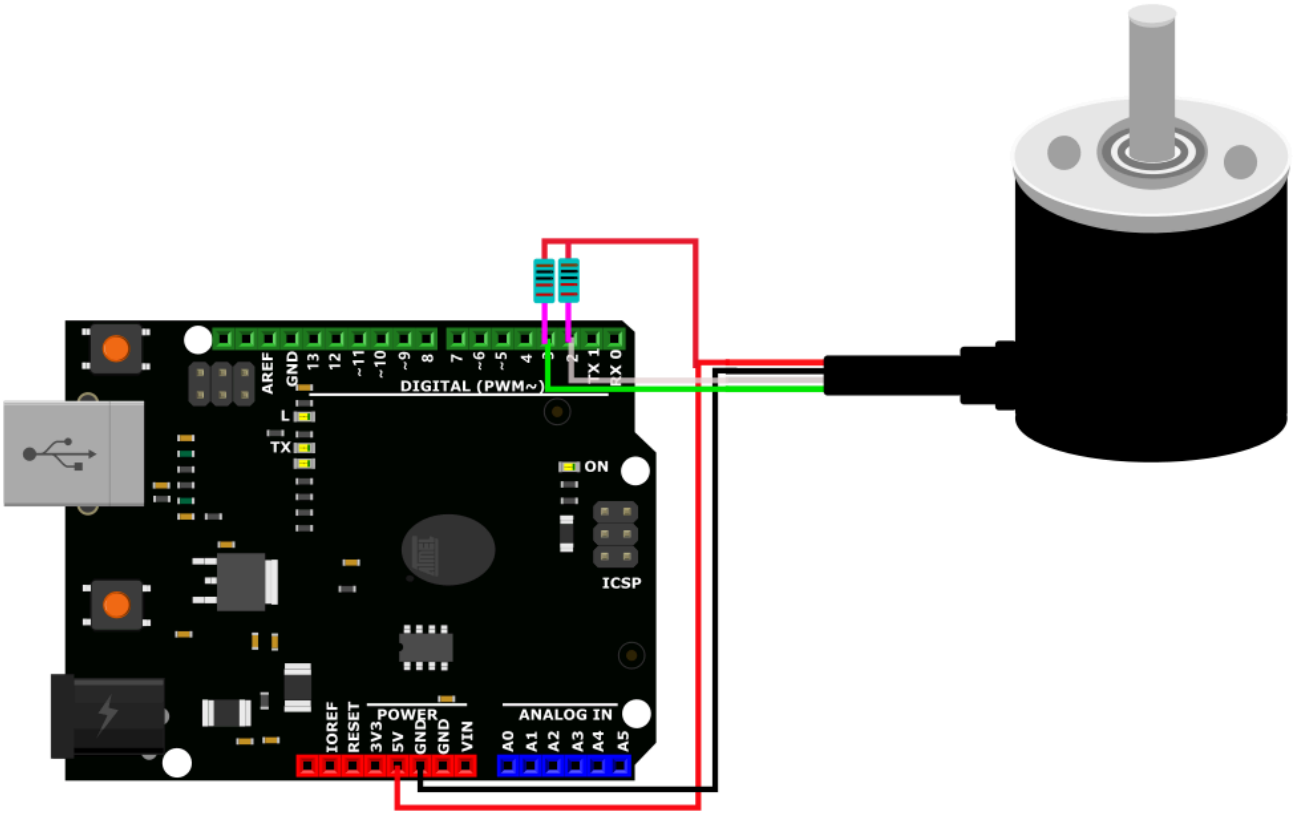
Tutorial

Direction & Interrupt count

Requirements

- **Hardware**
  - DFRduino UNO (or similar) x 1
  - Incremental Photoelectric Rotary Encoder
  - 2x 1K Resistor
  - M-M/F-M/F-F Jumper wires
- **Software**
  - Arduino IDE, Click to Download Arduino IDE from Arduino® (https://www.arduino.cc/en/software)

Connection Diagram

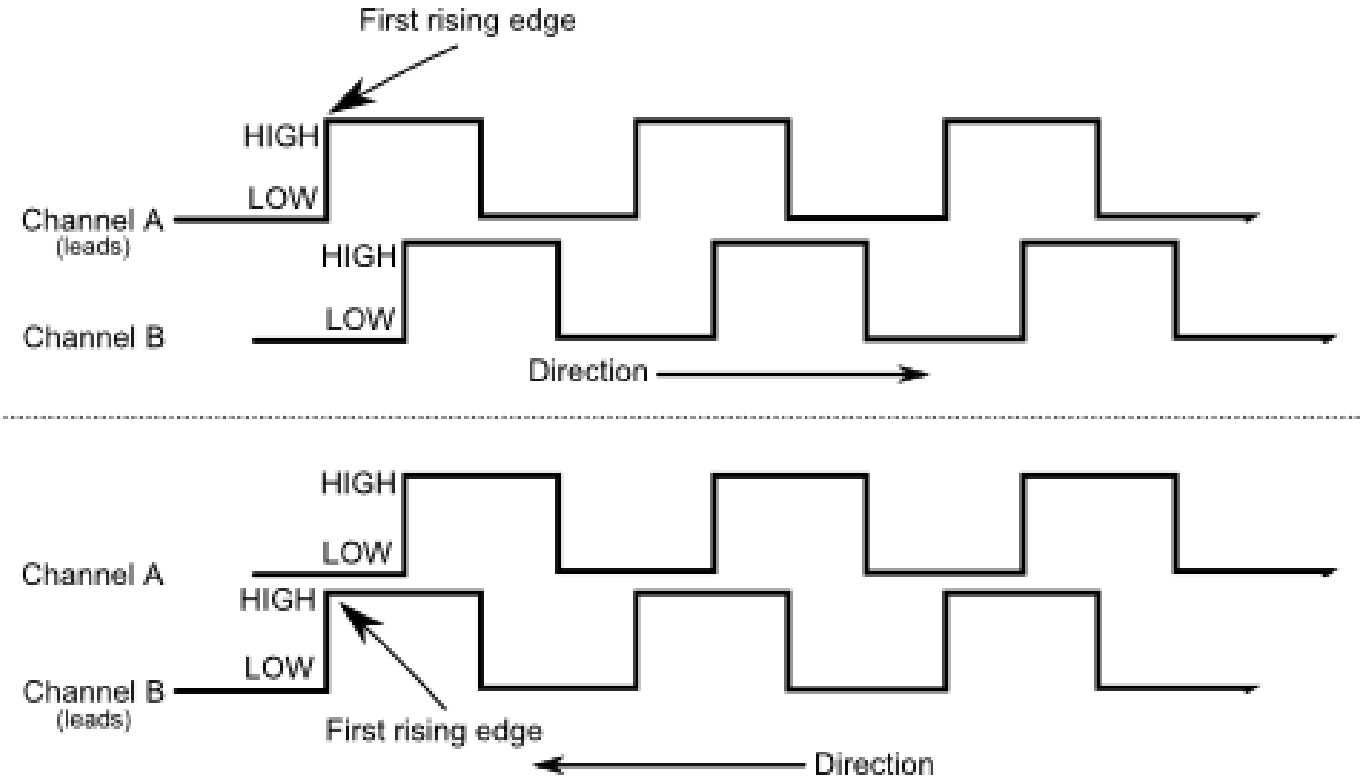


Sample Code

```
/*!
 * @file  SEN0230.ino
 * @brief  Two phase quadrature encoder(Incremental)
 * @n  To determine motor with encode （CW OR CCW）
 * @copyright  Copyright (c) 2010 DFRobot Co.Ltd (http://www.dfrobot.com)
 * @license  The MIT License (MIT)
 * @author  Dongzi(1185787528@qq.com)
 * @version  V1.0
 * @date  2016-5-26
 */

#define  A_PHASE 2
#define  B_PHASE 3
unsigned int flagA = 0;  //Assign a value to the token bit
unsigned int flagB = 0;  //Assign a value to the token bit
/** *  */
void setup() {
  pinMode(A_PHASE, INPUT);
  pinMode(B_PHASE, INPUT);
  Serial.begin(9600);   //Serial Port Baudrate: 9600
  attachInterrupt(digitalPinToInterrupt( A_PHASE), interrupt, RISING); //Inter
}
void loop() {

  Serial.print("CCW:  ");
  Serial.println(flagA);
  Serial.print("CW: ");
  Serial.println(flagB);
  delay(1000);// Direction judgement

}
void interrupt()// Interrupt function
{ char i;
  i = digitalRead( B_PHASE);
```

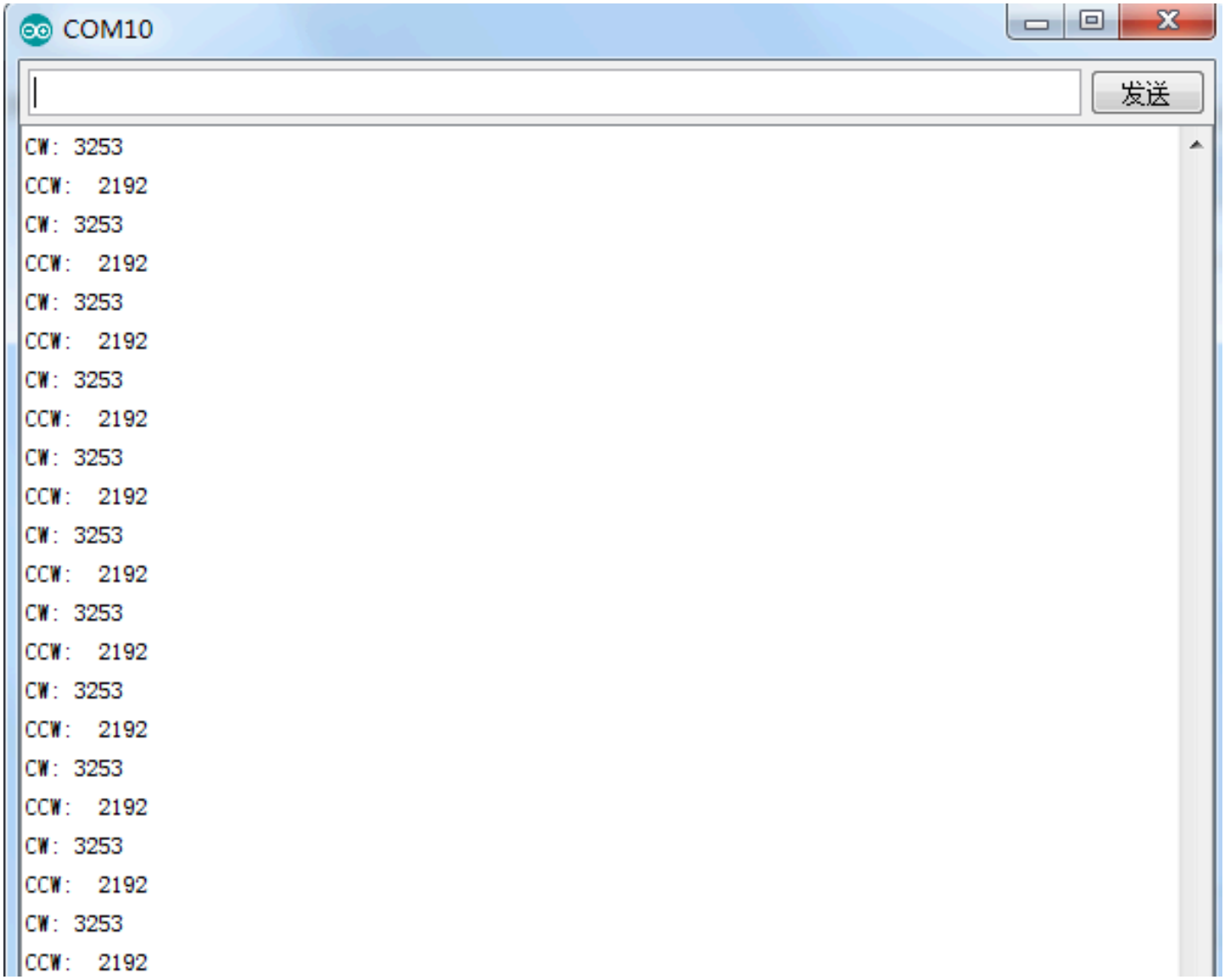
```
  if (i == 1)
    flagA += 1;
  else


    flagB += 1;
}
```

OUTPUT



Expected Results

Use the interruption to detect the rotation direction and count cylinder number.



FAQ

Q1. Why I can't get any feedback from the encoder?

A. Please connect pull-up resistor to phase A & B

For any questions, advice or cool ideas to share, please visit the DFRobot Forum (https://www.dfrobot.com/forum/).

More Documents